

# OMNIX

Decision Governance Infrastructure  
for Automated Systems

*"Preventing costly mistakes before they happen"*

Eureka! GCC 2026 — Semifinalist Report

Startup: OMNIX

Founder & CEO: Harold Nunes

Stage: MVP (Live Product)

Pre-Seed Round: \$500,000 USD

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# 1. Executive Summary

**Problem:** High-stakes automated decision systems — from trading to supply chain to lending — lack institutional-grade governance infrastructure, leading to billions in preventable losses annually.

**Solution:** OMNIX is Decision Governance Infrastructure that validates every automated decision through 6 independent checkpoints before execution, with a fail-closed architecture that defaults to protecting capital.

**Target Customer:** B2B — Prop trading firms, trading platforms, regulated funds, and family offices in ADGM/DIFC ecosystem (200+ registered firms). Future verticals: supply chain operators, lending platforms, insurance underwriters, and robotics/autonomous physical systems (Year 3+).

**Business Model:** B2B Decision Governance Licensing (\$15K–\$35K/month per enterprise client) + per-validation pricing (\$0.01–\$0.05/call) for high-volume platforms. Secondary B2C SaaS (\$149–\$499/month).

**Current Stage:** MVP (Live Product) — Running 24/7 in production since November 2025. 655,000+ evaluation cycles completed, 6,580+ cryptographically signed governance receipts, 100% post-quantum cryptography coverage.

**Key Traction:** Eureka GCC 2026 Semifinalist. Live public verification system at [omnibotgenesis-production.up.railway.app/verify](https://omnibotgenesis-production.up.railway.app/verify). Interactive governance demos for credit/lending and insurance underwriting verticals. Institutional website live at [www.omnixquantum.net](https://www.omnixquantum.net).

| Metric   | Description         | Context                                         |
|----------|---------------------|-------------------------------------------------|
| 655,000+ | Evaluation Cycles   | Governance engine operational 24/7              |
| 6,580+   | PQC-Signed Receipts | 100% Dilithium-3 coverage                       |
| 98.5%    | Capital Preserved   | During BTC -7.37% volatility                    |
| <120ms   | Decision Latency    | Real-time governance validation                 |
| \$49.7B+ | Combined TAM        | Multi-vertical addressable market (6 verticals) |

## 2. Company Overview

### Vision

To become the global standard for decision governance infrastructure in automated systems — ensuring that every high-stakes automated decision is validated, traceable, and accountable before execution.

### Business Description

OMNIX builds Decision Governance Infrastructure for automated systems. We provide a multi-layer pre-execution validation engine that governs decisions across domains where capital is at risk. Our 6-checkpoint architecture acts as an independent governance layer: it does not generate signals — it validates and governs them. The architecture is domain-agnostic, validated first in digital asset trading (the hardest domain due to 24/7 operation, high volatility, and zero circuit breakers), with expansion planned (Year 2–3+) into supply chain, credit/lending, insurance, energy trading, and robotics/autonomous systems.

### Product Description

OMNIX Decision Governance Engine is a real-time, AI-powered validation infrastructure that sits between decision systems and execution. Every decision passes through 6 independent checkpoints: Monte Carlo Validation, Risk Management System Limits, Adaptive Coherence Gate, Edge Confirmation Window, Weighted Scoring, and Final Decision. The system operates on a fail-closed principle — if any checkpoint raises concern, the decision is blocked. Every decision (executed or blocked) is cryptographically signed with NIST-standardized post-quantum algorithms (Dilithium-3) and stored with a full audit trail.

### Current Stage of Development

| Component                  | Status                                                                |
|----------------------------|-----------------------------------------------------------------------|
| Decision Governance Engine | Live — Production 24/7 since Nov 2025                                 |
| 6-Checkpoint Validation    | Live — All 6 checkpoints operational                                  |
| Post-Quantum Cryptography  | Live — Dilithium-3 + Kyber-768 since Nov 2025                         |
| Public Verification Server | Live — Public receipt verification endpoint                           |
| Investor Dashboard         | Live — 19 widgets, real-time metrics                                  |
| Institutional Website      | Live — <a href="http://www.omnixquantum.net">www.omnixquantum.net</a> |
| Multi-Vertical Demos       | Live — Credit/Lending + Insurance interactive demos                   |
| Shadow Portfolio Analysis  | Live — 192,000+ counterfactual events                                 |
| Enterprise API             | In Development — Q2 2026 target                                       |

### 3. Problem & Market Opportunity

#### Problem Statement

Automated decision systems across industries — algorithmic trading, supply chain procurement, credit approvals, insurance underwriting, and robotics/autonomous physical systems — operate without adequate pre-execution governance. The consequences are severe:

- **\$68B+ lost annually** by traders alone due to emotional and undisciplined decision-making
- **95% of algorithmic systems** ask 'When should I act?' instead of 'When should I NOT act?'
- **Single-layer risk checks** fail during tail events (flash crashes, liquidity cascades)
- **No audit trail** for automated decisions, creating regulatory and compliance gaps
- **Institutional-grade governance** exists only inside hedge funds with \$100M+ AUM — inaccessible to the rest of the market

#### Existing Alternatives & Why They Are Inadequate

| Alternative              | Limitation                                | OMNIX Advantage                        |
|--------------------------|-------------------------------------------|----------------------------------------|
| Retail Trading Bots      | No decision governance, single risk check | 6 independent checkpoint validation    |
| Quant Fund Tools         | Minimum \$10M+ capital, proprietary       | Accessible infrastructure model        |
| Manual Oversight         | Too slow for real-time (<120ms required)  | Automated validation in <120ms         |
| Single-Domain Tools      | Built for one industry only               | Domain-agnostic governance engine      |
| Traditional Risk Systems | React after the fact                      | Pre-execution validation (fail-closed) |

#### Market Size

OMNIX addresses a multi-vertical market opportunity across decision governance for automated systems:

| Vertical                        | Addressable Market                 | Timeline        |
|---------------------------------|------------------------------------|-----------------|
| Digital Asset Trading (Current) | \$18.8B algorithmic trading        | Now (Validated) |
| Supply Chain Risk               | \$3.2B supply chain analytics      | Year 2–3        |
| Credit / Lending Governance     | \$7.4B credit risk management      | Year 2–3        |
| Insurance Underwriting          | \$5.1B insurtech                   | Year 3+         |
| Energy Trading                  | \$2.8B energy risk management      | Year 3+         |
| Robotics / Autonomous Systems   | \$12.4B industrial robotics safety | Year 3+         |
| Combined TAM                    | \$49.7B+                           | Progressive     |

## Target Geography & Customer Segment

**Primary Geography:** ADGM/DIFC ecosystem (Abu Dhabi/Dubai) — 200+ registered prop trading firms, 100+ regulated funds, 50+ crypto-native platforms. **Expansion:** EU (MiCA compliance creating urgent demand), Asia-Pacific (Year 2–3).

**Primary Segment:** B2B Institutional (80% focus) — Prop trading firms, trading platforms, regulated funds, and family offices. **Secondary:** B2C SaaS for advanced independent traders (20%, post-enterprise validation).

## Expected Market Adoption

| Phase             | Timeline    | Target                                    |
|-------------------|-------------|-------------------------------------------|
| Enterprise Pilots | Months 1–6  | 3 pilot agreements (prop firms/platforms) |
| Paid Licenses     | Months 6–9  | 3 paying enterprise clients               |
| Regional Scale    | Months 9–12 | \$50K+ MRR, 15–30 clients in MENA         |
| Multi-Vertical    | Year 2–3    | Supply chain + lending governance pilots  |
| Global Scale      | Year 3+     | EU MiCA, Asia expansion                   |

## 4. Customer Validation & Insights

### Discovery Method

Customer validation has been conducted through multiple channels: direct conversations with prop trading firm managers in the ADGM/DIFC ecosystem, participation in the Eureka GCC competition (semifinalist status achieved), institutional investor due diligence sessions, and live product demonstrations with the governance engine running in real-time production.

### Key Insights Learned

- **Governance, not alpha:** Institutional buyers care about decision accountability and audit trails, not signal generation
- **Compliance urgency:** MiCA regulation (2025+) creates immediate demand for decision governance documentation
- **Risk-first positioning resonates:** 'Preventing costly mistakes' is a stronger value proposition than 'maximizing returns'
- **Fail-closed architecture is differentiating:** Institutions value systems that default to protect over systems that default to act
- **Multi-vertical potential recognized:** The same governance engine logic applies to credit, insurance, supply chain, and robotics/autonomous systems decisions

### Willingness to Pay

Based on market research and competitive analysis of institutional risk infrastructure pricing:

| Segment              | Expected Price Range      | Basis                                                           |
|----------------------|---------------------------|-----------------------------------------------------------------|
| Prop Trading Firms   | \$10K–\$15K/month (pilot) | Comparable to existing risk tools; drawdown prevention ROI 2–8x |
| Trading Platforms    | \$0.01–\$0.05/validation  | Per-API-call model; volume-driven                               |
| Regulated Funds      | \$25K–\$35K/month         | Audit trail + compliance documentation value                    |
| B2C Advanced Traders | \$149–\$499/month         | SaaS market benchmarks for institutional tools                  |

### Customer Testimonials

[Section reserved for 5 customer testimonials with name, role, contact details, and key feedback. Testimonials are being collected from pilot conversations and will be included in the final submission.]

| # | Name          | Role   | Contact       | Key Feedback                                        |
|---|---------------|--------|---------------|-----------------------------------------------------|
| 1 | [To be added] | [Role] | [Email/Phone] | [Pain point, value perception, pricing sensitivity] |

| # | Name          | Role   | Contact       | Key Feedback                                        |
|---|---------------|--------|---------------|-----------------------------------------------------|
| 2 | [To be added] | [Role] | [Email/Phone] | [Pain point, value perception, pricing sensitivity] |
| 3 | [To be added] | [Role] | [Email/Phone] | [Pain point, value perception, pricing sensitivity] |
| 4 | [To be added] | [Role] | [Email/Phone] | [Pain point, value perception, pricing sensitivity] |
| 5 | [To be added] | [Role] | [Email/Phone] | [Pain point, value perception, pricing sensitivity] |



## 5. Product & Technology

### Core Features Currently Built

| Feature                            | Status | Description                                                                                                        |
|------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------|
| 6-Checkpoint Governance Engine     | Live   | Multi-layer pre-execution decision validation with independent veto authority per checkpoint                       |
| Monte Carlo Validation             | Live   | 10,000 simulation paths per decision; blocks trades with negative expected return                                  |
| Adaptive Coherence Gate            | Live   | Dynamically calibrated signal agreement scoring based on market regime severity                                    |
| Edge Confirmation Window (ECW)     | Live   | Requires 3 consecutive cycles of confirmed statistical edge before execution                                       |
| Black Swan Detection               | Live   | Real-time tail risk monitoring with automatic exposure reduction (4 severity levels)                               |
| Decision Contradiction Index (DCI) | Live   | Measures internal signal divergence (0–100); high DCI mandates HOLD                                                |
| Post-Quantum Cryptography          | Live   | NIST-standardized Dilithium-3 signatures + Kyber-768 key encapsulation                                             |
| Public Verification Server         | Live   | Public receipt verification endpoint with zero internal data exposure                                              |
| Shadow Portfolio Engine            | Live   | 192,000+ counterfactual events; validates veto accuracy (100% accuracy on 50 validated)                            |
| Non-Markovian Memory Kernel        | Live   | Behavioral pattern detection beyond recency bias                                                                   |
| Multi-AI Orchestration             | Live   | Gemini 2.5 Flash + GPT-4o + Claude Sonnet 4 with automatic failover                                                |
| Investor Dashboard                 | Live   | 19 widgets with real-time metrics, dual win rate framework, regime detection                                       |
| Institutional Website              | Live   | Public landing at <a href="http://www.omnixquantum.net">www.omnixquantum.net</a> with live market data integration |
| Interactive Governance Demos       | Live   | Credit/Lending + Insurance underwriting demos showing multi-vertical applicability                                 |

### What Is Live vs What Is Planned

| Component                          | Status | Timeline       |
|------------------------------------|--------|----------------|
| Governance Engine (Digital Assets) | LIVE   | Since Nov 2025 |

| Component                             | Status  | Timeline       |
|---------------------------------------|---------|----------------|
| PQC Decision Signing                  | LIVE    | Since Nov 2025 |
| Public Verification System            | LIVE    | Since Feb 2026 |
| Enterprise API (Risk Guardian)        | Planned | Q2 2026        |
| White-Label SDK                       | Planned | Q3 2026        |
| Supply Chain Domain Adapter           | Planned | Year 2–3       |
| Credit/Lending Domain Adapter         | Planned | Year 2–3       |
| Insurance Domain Adapter              | Planned | Year 3+        |
| Robotics / Autonomous Systems Adapter | Planned | Year 3+        |

### Product Roadmap (Next 6–12 Months)

- **Q1 2026:** Complete track record validation. Finalize institutional documentation. Eureka GCC competition.
- **Q2 2026:** Enterprise API launch (Risk Guardian API). First enterprise pilot (prop firm or trading platform). ADGM regulatory structure initiated.
- **Q3 2026:** Public license model launch. White-label SDK for platform integrations. Second and third enterprise clients.
- **Q4 2026:** Series A readiness with validated revenue metrics. Begin multi-vertical domain adapter development.

### Technology Stack

| Layer          | Technology                                                        |
|----------------|-------------------------------------------------------------------|
| Language       | Python 3.11 (core engine)                                         |
| Frontend       | React 18 + TypeScript + Tailwind CSS + Vite                       |
| Database       | PostgreSQL (42+ tables, 90% FK coverage)                          |
| Cache          | Redis (state management, rate limiting)                           |
| AI Models      | Google Gemini 2.5 Flash, OpenAI GPT-4o, Anthropic Claude Sonnet 4 |
| Cryptography   | CRYSTALS-Dilithium-3 (ML-DSA-65), CRYSTALS-Kyber-768 (ML-KEM-768) |
| Infrastructure | Railway (production 24/7), Replit (development)                   |
| Verification   | aiohttp async server (port 8000), SHA-256 hash chain              |
| Architecture   | Hexagonal (ports and adapters), 27 Architecture Decision Records  |

### Key Technical or Execution Risks

- **Key-person dependency:** Solo founder. Mitigated by documented hexagonal architecture (27 ADRs, 2–3 week onboarding target). First 3 hires planned within Month 1–4 post-funding.

- **AI model dependency:** Reliance on third-party AI providers. Mitigated by multi-provider failover chain (Gemini → GPT-4o → Claude).
- **Market timing:** Enterprise sales cycles are 3–6 months. Conservative pipeline assumptions account for this.

**Demo Links:** Public verification: [omnibotgenesis-production.up.railway.app/verify](https://omnibotgenesis-production.up.railway.app/verify) | Website: [www.omnixquantum.net](https://www.omnixquantum.net) | Credit governance demo: [www.omnixquantum.net/governance-demo](https://www.omnixquantum.net/governance-demo) | Insurance governance demo: [www.omnixquantum.net/governance-demo-insurance](https://www.omnixquantum.net/governance-demo-insurance)

## 6. Competitive Landscape & Differentiation

OMNIX operates in the decision governance infrastructure space, which is a new category. The closest comparisons come from adjacent markets:

| Competitor           | Category            | Decision Checkpoints | PQC Security | Multi-Vertical     | Pricing            |
|----------------------|---------------------|----------------------|--------------|--------------------|--------------------|
| 3Commas              | Trading Platform    | 1 (basic SL/TP)      | No           | No                 | \$49–\$79/mo       |
| Gauntlet Networks    | DeFi Risk           | 2–3                  | No           | DeFi only          | Enterprise         |
| Chainalysis          | Compliance          | 0 (monitoring only)  | No           | Compliance only    | \$100K+/yr         |
| Internal Quant Teams | Prop Funds          | 2–4                  | Rare         | No                 | \$1M+/yr team cost |
| OMNIX                | Decision Governance | 6 independent        | Yes (NIST)   | Yes (6+ verticals) | \$15K–\$35K/mo     |

### Unique Selling Proposition (USP)

OMNIX is the first domain-agnostic Decision Governance Infrastructure with production-integrated post-quantum cryptography, 6 independent validation checkpoints, and a publicly verifiable decision audit trail. No competitor offers this combination.

### Why We Win

#### Today:

- **Fail-closed by default:** Every other system defaults to execute. OMNIX defaults to protect.
- **Post-quantum security in production:** No competitor has PQC-signed decision receipts.
- **Full audit trail:** 6,580+ governance receipts, publicly verifiable, zero information leakage.
- **Domain-agnostic architecture:** Same engine validates trading, credit, insurance, and robotics/autonomous systems decisions.

#### At Scale:

- **Network effects:** More decisions governed = better calibration. Shadow Portfolio learns from every veto.
- **Multi-vertical expansion:** One architecture, multiple verticals (trading, supply chain, lending, insurance, energy, robotics). Each new domain increases TAM without rebuilding core.
- **Regulatory moat:** MiCA, ADGM, and emerging compliance requirements create demand for auditable decision governance.
- **Switching costs:** OMNIX embeds into execution flow — high switching cost once integrated.

# 7. Business Model & Unit Economics

## Revenue Model

OMNIX operates a dual revenue model: B2B Decision Governance Licensing (80% of revenue) and B2C SaaS (20%, post-enterprise validation).

| Product            | Pricing                     | Target Customer                       |
|--------------------|-----------------------------|---------------------------------------|
| Risk Guardian API  | \$15K–\$35K/month           | Prop firms, trading platforms         |
| White-Label Engine | \$100K+ setup + \$20K/month | Exchanges, brokers                    |
| Per-Validation API | \$0.01–\$0.05/call          | High-volume platforms                 |
| B2C Pro            | \$149/month                 | Advanced independent traders          |
| B2C Advanced       | \$499/month                 | Semi-professional traders, API access |

## Pricing Strategy

Pricing is positioned below the cost of building proprietary risk governance internally (\$1M+ annually for a quant team) while delivering institutional-grade protection. Enterprise pilots begin at discounted rates (\$10K–\$15K/month) with conversion to full pricing after validation period.

*"Institutions pay for what blocks bad decisions, not for alpha. License-based revenue. No tokens. No performance fees."*

## Customer Acquisition Channels

- **Direct outreach:** 50 targeted contacts/month via LinkedIn + email (targeting 15 meetings/month)
- **ADGM Innovation Hub:** Regulatory community + startup programs (5–10 warm intros/quarter)
- **Hub71:** Applied — pending response (3–5 qualified leads/quarter if accepted)
- **Industry events:** TOKEN2049 Dubai, FinTech Abu Dhabi, GITEX (10+ contacts/event)
- **Eureka GCC:** Competition exposure + judge network (immediate pipeline)

## Unit Economics (Estimated)

| Metric           | Value            | Basis                                         |
|------------------|------------------|-----------------------------------------------|
| CAC (Enterprise) | \$5,000–\$10,000 | Direct sales, 3–6 month cycle, founder-led    |
| LTV (Enterprise) | \$180K–\$420K    | \$15K–\$35K/month × 12-month minimum contract |
| LTV/CAC Ratio    | 18x–84x          | Strong unit economics at enterprise tier      |
| CAC (B2C SaaS)   | \$50–\$150       | Content marketing + referral programs         |
| LTV (B2C SaaS)   | \$1,788–\$5,988  | \$149–\$499/month × 12-month avg retention    |

| Metric       | Value        | Basis                     |
|--------------|--------------|---------------------------|
| Gross Margin | 60–70%       | SaaS infrastructure model |
| Break-Even   | 18–24 months | Conservative estimate     |

## 8. Traction & Key Metrics

OMNIX has been running in production 24/7 since November 2025. The following metrics represent real system telemetry, not projections:

| Metric                 | Value               | Significance                                              |
|------------------------|---------------------|-----------------------------------------------------------|
| Production Uptime      | 24/7 since Nov 2025 | System operational for 4+ months continuously             |
| Evaluation Cycles      | 655,000+            | Governance engine processing decisions in real-time       |
| PQC-Signed Receipts    | 6,580+              | Every decision signed with Dilithium-3 (100% coverage)    |
| Capital Preserved      | 98.5%               | During period when BTC dropped 7.37%                      |
| Shadow Trade Events    | 192,000+            | Counterfactual analysis of vetoed decisions               |
| Veto Accuracy          | 100%                | 50 validated shadow outcomes — all vetos were correct     |
| Decision Latency       | <120ms              | Real-time governance validation                           |
| Database Tables        | 42+                 | 90% foreign key coverage — institutional-grade data model |
| Architecture Decisions | 27 ADRs             | Documented engineering discipline                         |
| Dashboard Widgets      | 19/19               | Full operational visibility                               |

## Competition & Recognition

- **Eureka GCC 2026:** Semifinalist — competing in Abu Dhabi startup competition
- **Public verification:** Live at [omnibotgenesis-production.up.railway.app/verify](https://omnibotgenesis-production.up.railway.app/verify)
- **Institutional website:** Live at [www.omnixquantum.net](https://www.omnixquantum.net) with real-time market data

## Key Learning Milestones

| Date         | Milestone                                 | Impact                                         |
|--------------|-------------------------------------------|------------------------------------------------|
| Nov 2025     | PQC implemented (Dilithium-3 + Kyber-768) | Quantum-resistant decision signing operational |
| Nov–Dec 2025 | Learning Baseline (119 trades)            | System calibration, threshold discovery        |
| Jan 15, 2026 | Official Track Record Day 1               | Clean metrics from calibrated parameters       |
| Jan 21, 2026 | Edge Confirmation Window + DCI            | Capital patience + contradiction detection     |
| Jan 28, 2026 | Institutional Website launched            | Public credibility and demo capability         |

| Date         | Milestone                                 | Impact                                    |
|--------------|-------------------------------------------|-------------------------------------------|
| Feb 15, 2026 | Multi-Vertical Demos (Credit + Insurance) | Domain-agnostic architecture demonstrated |
| Feb 21, 2026 | Public Verification Server                | Transparent governance receipts           |
| Feb 23, 2026 | Railway cost optimization (77% reduction) | Operational efficiency for scaling        |

Note: All metrics are from internal dataset, not externally audited. Evaluation cycles and receipts represent governance engine activity, not trading performance.



## 9. Go-To-Market Strategy

### Customer Acquisition Strategy

OMNIX follows an enterprise-first go-to-market strategy, targeting institutional buyers in the ADGM/DIFC ecosystem where regulatory urgency (MiCA 2025+) creates immediate demand for decision governance infrastructure.

| Stage         | Action                                           | Timeline  | Conv. Rate |
|---------------|--------------------------------------------------|-----------|------------|
| Awareness     | 50 direct outreaches/month + event networking    | Ongoing   | —          |
| Assessment    | Free governance health check for their system    | Week 1–2  | 30%        |
| Shadow Mode   | OMNIX runs alongside their system (no execution) | 2–4 weeks | 60%        |
| Advisory Mode | OMNIX provides pre-execution recommendations     | 2–4 weeks | 75%        |
| Enforcement   | Full governance integration with veto authority  | Ongoing   | 80%        |
| Paid License  | Monthly enterprise license                       | Month 3–6 | —          |

### Sales Cycle

Enterprise sales cycle: 3–6 months from first contact to paid license. The progressive onboarding model (Shadow → Advisory → Enforcement) reduces adoption risk and builds trust before requiring financial commitment. Founder-led sales during pre-seed phase.

### Key Partnerships (Target)

| Partner Type         | Target                         | Status                     |
|----------------------|--------------------------------|----------------------------|
| Regulatory Ecosystem | ADGM (Abu Dhabi)               | Target licensing ecosystem |
| Accelerator          | Hub71                          | Applied — pending response |
| Future Expansion     | DIFC (Dubai)                   | Roadmap                    |
| Technology           | Cloud infrastructure (Railway) | Active                     |
| AI Providers         | Google, OpenAI, Anthropic      | Active (multi-provider)    |

### Scaling Strategy Across Markets

| Phase   | Focus                                                                         | Timeline     |
|---------|-------------------------------------------------------------------------------|--------------|
| Phase 1 | Pilot validation — 3 enterprise clients (digital assets)                      | Months 1–6   |
| Phase 2 | Regional expansion (ADGM/DIFC/MENA) + platform partnerships                   | Months 6–12  |
| Phase 3 | Multi-vertical expansion — supply chain, lending, robotics/autonomous systems | Months 12–24 |
| Phase 4 | Global scaling (EU MiCA, Asia) + insurance, energy                            | Months 24–36 |

# 10. Team & Advisors

## Founder

**Harold Nunes** — Founder & CEO

- Product architecture, risk logic design, and infrastructure development
- Built OMNIX end-to-end: from concept to production system running 24/7
- AI-augmented development methodology: one person with AI achieves the output of a 5-person team
- Deep domain expertise in algorithmic trading risk management, decision systems, and cryptographic security
- Designed the 6-checkpoint governance architecture, post-quantum integration, and multi-vertical domain adapter pattern

## Key Hires Planned (Post-Funding)

| Role                    | Timeline  | Purpose                                        |
|-------------------------|-----------|------------------------------------------------|
| Senior Backend Engineer | Month 1–2 | Enterprise API development, system scaling     |
| DevOps / Infrastructure | Month 2–3 | Production reliability, deployment automation  |
| Business Development    | Month 3–4 | Enterprise sales, ADGM ecosystem relationships |

## Key-Person Risk Mitigation

The current key-person risk is acknowledged and actively mitigated:

- **Documented architecture:** 27 Architecture Decision Records (ADRs), hexagonal architecture with clear module boundaries
- **Onboarding target:** 2–3 week onboarding for new engineers using existing documentation
- **Dependency reduction:** First 3 hires reduce founder dependency from 100% to ~30% by Month 4
- **IP protection:** IP assignment to company entity, key-person insurance, operational runbooks by Month 6

## Advisors

Advisory board is being assembled. The founder has worked with contract developers and infrastructure consultants during the development phase. Formal advisory relationships with ADGM ecosystem participants and fintech domain experts are targeted post-funding.

## Why This Team

OMNIX was built by a founder who understands both the technical complexity of multi-layer governance systems and the institutional requirements of regulated financial markets. The AI-augmented development model has enabled solo construction of a system that would typically require a 5-person engineering team — evidenced by 42+ database tables, 27 ADRs, 655,000+ evaluation cycles, and 4+ months of continuous production operation.

Post-funding, the first hires are designed to eliminate key-person risk and accelerate enterprise sales.

# 11. Financials & Fundraising

## 3-Year Financial Projections

|                    | Year 1            | Year 2            | Year 3                      |
|--------------------|-------------------|-------------------|-----------------------------|
| Enterprise Clients | 3 pilots          | 5–8 licenses      | 15–30 licenses              |
| B2C SaaS Users     | 50–100            | 500–1,000         | 2,000–5,000                 |
| Revenue            | \$200K–\$400K     | \$800K–\$1.2M     | \$2M+                       |
| Gross Margin       | 60–70%            | 65–75%            | 70–80%                      |
| Key Driver         | Enterprise pilots | License expansion | Multi-vertical + SaaS scale |

## Key Assumptions

- 3 enterprise pilots secured within 6 months via ADGM/DIFC ecosystem
- Average enterprise contract: \$25K/month (mid-range of \$15K–\$35K pricing)
- 12-month minimum contract duration with 80% renewal rate
- B2C SaaS launched post-enterprise validation (Month 6–9)
- Infrastructure costs scale linearly with client count at 60–70% gross margin
- MiCA compliance urgency drives accelerated adoption in EU markets (Year 2+)

## Fundraising

| Item                 | Details                 |
|----------------------|-------------------------|
| Raising              | \$500,000 USD           |
| Equity               | 16.7%                   |
| Pre-Money Valuation  | \$2.5M–\$3M             |
| Funds Raised to Date | \$0 (bootstrapped)      |
| Instrument           | Equity (pre-seed round) |

## Valuation Justification

- **Working product in production:** 4+ months running 24/7 with real market data
- **Real validation data:** 655,000+ evaluation cycles, 6,580+ PQC-signed receipts
- **Defensible IP:** 6-checkpoint architecture + Shadow Portfolio engine + PQC integration
- **Strategic timing:** MiCA + ADGM convergence creating urgent demand for governance infrastructure
- **Comparable:** Chainalysis raised at \$4M pre-money at similar stage

## Use of Funds

| Category                      | Allocation | Purpose                                                          |
|-------------------------------|------------|------------------------------------------------------------------|
| Strategy & Risk Engine        | 35%        | Algorithm refinement, Shadow Portfolio expansion, enterprise API |
| Dubai/ADGM Legal & Regulatory | 25%        | Company formation, regulatory structure, compliance              |
| Enterprise Infrastructure     | 20%        | API for prop firms, security certifications, scaling             |
| Team & Operations             | 15%        | 2–3 key hires eliminating key-person risk (Month 1–4)            |
| Reserve                       | 5%         | Contingency                                                      |

## 12. Intellectual Property & Regulatory

### Intellectual Property

OMNIX's intellectual property is embedded in its proprietary architecture and algorithms, not in traditional patents. The key defensible assets are:

| IP Asset                       | Description                                                | Protection                        |
|--------------------------------|------------------------------------------------------------|-----------------------------------|
| 6-Checkpoint Governance Engine | Multi-layer pre-execution decision validation architecture | Trade secret + documented ADRs    |
| Adaptive Coherence Gate        | Dynamic threshold calibration based on market regime       | Proprietary algorithm             |
| Edge Confirmation Window       | Statistical edge persistence validation (3-cycle)          | Proprietary methodology           |
| Decision Contradiction Index   | Internal signal divergence measurement (0–100)             | Proprietary metric                |
| Shadow Portfolio Engine        | 192,000+ counterfactual events for veto validation         | Proprietary dataset + methodology |
| Non-Markovian Memory Kernel    | Behavioral pattern detection beyond recency bias           | Proprietary algorithm             |
| PQC Integration Architecture   | Production-integrated post-quantum decision signing        | First-mover advantage             |

Patent filing strategy will be evaluated post-funding with legal counsel in ADGM jurisdiction. Current protection relies on trade secrets, first-mover advantage, and architectural complexity.

### Regulatory Framework

| Framework        | Status           | Relevance                                                 |
|------------------|------------------|-----------------------------------------------------------|
| ADGM (Abu Dhabi) | Target ecosystem | Company formation + licensing structure                   |
| MiCA (EU)        | Creating demand  | Decision governance documentation required for compliance |
| GDPR             | Controls aligned | Data protection and PII handling                          |
| SOC 2 Principles | Aligned          | Access control, audit logging, encryption                 |
| MiFID II         | Aligned          | Decision auditability and traceability                    |
| DORA             | Aligned          | Operational resilience and error isolation                |

### Compliance Risks & Mitigation

- **Regulatory classification:** OMNIX is infrastructure (not a financial product or fund). No asset custody, no investment advice, no token issuance.
- **Data residency:** PostgreSQL database with configurable deployment region. ADGM/DIFC data handling requirements addressed through cloud infrastructure selection.
- **Cryptographic compliance:** NIST-standardized algorithms (Dilithium-3, Kyber-768). No proprietary or unvetted cryptographic implementations.



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## 13. Risks & Challenges

### Top 3 Business Risks

#### 1. Key-Person Risk (HIGH)

OMNIX is currently built and operated by a solo founder. This creates dependency risk for investors and operational continuity.

**Mitigation:** Documented architecture (27 ADRs), hexagonal module design with clear boundaries, 2–3 week onboarding target. First 3 hires (Month 1–4) reduce dependency from 100% to ~30%. IP assignment, key-person insurance, and operational runbooks by Month 6.

#### 2. Enterprise Sales Cycle Risk (MEDIUM)

Enterprise sales cycles are typically 3–6 months. Delayed pilot agreements could impact runway and milestone timelines.

**Mitigation:** Progressive onboarding model (Shadow → Advisory → Enforcement) reduces adoption friction. ADGM/DIFC ecosystem provides warm introductions. MiCA compliance urgency accelerates decision-making. \$500K runway provides 18–24 month operating buffer.

#### 3. Market Timing Risk (MEDIUM)

Crypto market volatility could reduce institutional appetite for digital asset infrastructure during bear markets.

**Mitigation:** OMNIX is positioned as decision governance infrastructure, not a crypto product. The multi-vertical architecture (supply chain, lending, insurance, robotics/autonomous systems) reduces dependency on any single market. Bear markets actually increase demand for risk governance.

### Execution Challenges

- **Scaling from solo to team:** First hires must maintain code quality and architectural discipline. Hexagonal architecture and ADR documentation mitigate this.
- **Enterprise integration complexity:** Each client requires customized onboarding. Domain Adapter pattern standardizes this process.
- **AI provider reliability:** Multi-provider failover chain (Gemini → GPT-4o → Claude) ensures no single point of failure.

### Market & Regulatory Risks

- **Regulatory changes:** ADGM/DIFC regulations could shift. OMNIX's compliance-aligned architecture makes adaptation straightforward.
- **Competition from incumbents:** Large exchanges could build in-house governance. OMNIX's 4-month head start, PQC integration, and multi-vertical design (6+ verticals including robotics) create defensible position.
- **AI regulation:** Emerging AI governance regulations (EU AI Act) could affect multi-AI orchestration. OMNIX's transparent decision audit trail is aligned with explainability requirements.

# 14. Appendix

## A. Architecture Overview

|                                                                      |
|----------------------------------------------------------------------|
| Decision Engine                                                      |
|                                                                      |
| +-- Signal Detection (EMA Regime, HMM, Kalman, Non-Markovian, Kelly) |
|                                                                      |
| +-- 6-Checkpoint Governance Flow:                                    |
| 1. Monte Carlo VETO (10,000 simulations)                             |
| 2. RMS VETO (VaR95, drawdown limits)                                 |
| 3. Adaptive Coherence Gate (dynamic thresholds)                      |
| 4. Edge Confirmation Window (3-cycle persistence)                    |
| 5. Weighted Scoring (5 inputs, 100 points)                           |
| 6. Final Decision (EXECUTE / HOLD / BLOCK)                           |
|                                                                      |
| +-- Post-Decision:                                                   |
| +-- Dilithium-3 PQC Signature                                        |
| +-- SHA-256 Hash Chain                                               |
| +-- PostgreSQL Persistence                                           |
| +-- Public Verification Endpoint                                     |

## B. Scoring Logic

| Input                | Weight | Max Points           |
|----------------------|--------|----------------------|
| EMA Regime Signal    | 40%    | 40 pts               |
| HMM Regime Detection | 25%    | 25 pts               |
| Kalman Filter        | 15%    | 15 pts               |
| Non-Markovian Memory | 15%    | 15 pts               |
| Kelly Criterion      | 10%    | 10 pts (sizing only) |

Separate Veto/Penalty layer (Monte Carlo, Black Swan, Sentiment, Quantum Momentum) applies only penalties — it cannot add positive score.

## C. Decision Receipt Example

```
{  
  
  "receipt_id": "RCP-2026022115...",  
  
  "timestamp": "2026-02-21T23:55:39Z",  
  
  "asset": "BTC/USD",  
  
  "decision": "HOLD",  
  
  "content_hash": "sha256:a7f3e2...",  
  
  "prev_hash": "sha256:8b4c1d... (truncated)",  
  
  "signature_algorithm": "Dilithium-3 (ML-DSA-65)",  
  
  "governance_gates": [  
  
    "MC_VETO", "RMS_CHECK", "COHERENCE_GATE",  
  
    "ECW_GATE", "SCORING", "FINAL_DECISION"  
  
  ]  
  
}
```

## D. Links & Resources

- **Website:** [www.omnixquantum.net](http://www.omnixquantum.net)
- **Public Verification:** [omnibotgenesis-production.up.railway.app/verify](https://omnibotgenesis-production.up.railway.app/verify)
- **Credit Governance Demo:** [www.omnixquantum.net/governance-demo](http://www.omnixquantum.net/governance-demo)
- **Insurance Governance Demo:** [www.omnixquantum.net/governance-demo-insurance](http://www.omnixquantum.net/governance-demo-insurance)
- **LinkedIn:** [linkedin.com/in/harold-nunes-21bb65285](https://linkedin.com/in/harold-nunes-21bb65285)
- **Contact:** [contacto@omnixquantum.net](mailto:contacto@omnixquantum.net)

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**OMNIX — Decision Governance Infrastructure for Automated Systems**  
*"When systems can act, but choose discipline."*

Eureka GCC 2026 — Semifinalist | Abu Dhabi, UAE  
All metrics from internal dataset, not externally audited. Past performance does not guarantee future results.